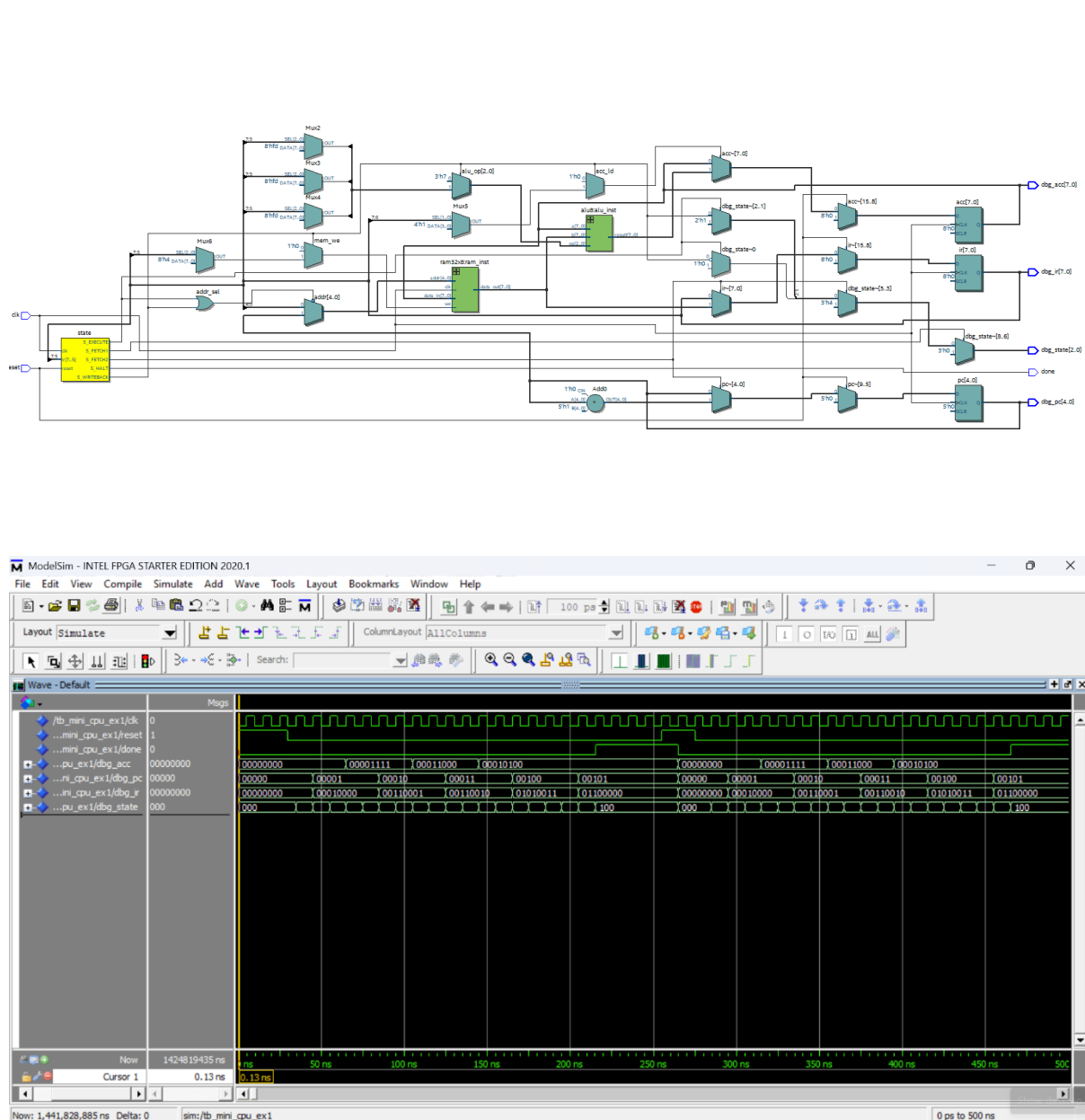


Washu2-CPU Week-5 Report

HARSHA VARDHAN

mini_cpu:



1. Initially, the instruction register was loading incorrect values because of RAM read latency. This was fixed by separating instruction fetch into S_FETCH1 and S_FETCH2, allowing one full clock cycle before asserting ir_id.
2. The accumulator was updating using stale memory data during execution. Moving acc_id assertion from S_EXECUTE to S_WRITEBACK ensured that mem_data_out became valid before loading the result.
3. Waveform verification using debug signals (dbg_acc, dbg_pc, dbg_ir, dbg_state) helped confirm correct state transitions, instruction decoding, and final result generation (ACC = 0x14, done = '1').